

Final

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Set up

[Github repo](https://github.com/jmagner4/ENVS-193DS_spring-2026_final): https://github.com/jmagner4/ENVS-193DS_spring-2026_final

Read in packages

```
# read in packages
library(tidyverse) # general use
library(here) # file organization
library(janitor) # cleaning data frames
library(ggeffects) # getting model predictions
library(MuMIn) # for AICc
library(DHARMA) # diagnostic plots

# store data in new object
nests<- read_csv(here("data","nest_data_final.csv"))

my_data <- read_csv(here("data","personaldatapoint.csv"))
```

Problems

Problem 1. Research Writing

a. Transparent statistical methods

In part 1, I think they used binomial/logistic regression. If habitat use is measured as used/not used, then the response variable would be binary. In part 2, I think they used a Spearman's

correlation test. Wanting to know the relationship between two variables suggests a correlation test should be used. Since both variables are categorical/nominal that suggests a non-parametric test should be used.

b. Suggetions for Rewriting

Part 1: Our evidence suggests that the distance to the water's edge is an important factor in determining where American Avocet choose to live.

Part 2: Different shorebird species, including American Avocet, Forster's Tern, Black-necked Stilt, prefer different types of coastal habitats.

c. Further Analysis

One potential variable influencing the probability of American Avocet microhabitat use is the availability of food or prey species in the habitats as well; this population would be a discrete variable (measured as a number of individuals per one-meter net sweep for example). I expect American Avocets to choose areas with higher prey populations because it makes finding food and surviving easier.

Problem 2. Data Analysis

a. Clean and Wrangle Data

```
# create new object for cleaned data
nests_clean <- nests |>

# select columns of interest
select(ant_species, case_control, height_cm)|>

# include only 3 ants species of interest
filter(ant_species %in% c("AB", "BBR", "RRB"))|>

# factor reorders levels of ant species as directed
mutate(
  ant_species = factor(ant_species, levels = c("AB", "RRB", "BBR")),

# include new column of full scientific names
  full_scientific_name = case_when(
    ant_species == "AB" ~ "Crematogaster sjostedti",
```

```

    ant_species == "RRB" ~ "Crematogaster mimosae",
    ant_species == "BBR" ~ "Crematogaster nigriceps"))

# display structure
str(nests_clean)

```

```

tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ ant_species      : Factor w/ 3 levels "AB","RRB","BBR": 2 2 2 2 1 2 3 2 1 3 ...
 $ case_control     : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm        : num [1:270] 392 310 513 154 324 ...
 $ full_scientific_name: chr [1:270] "Crematogaster mimosae" "Crematogaster mimosae" "Cremat

```

```

# display 10 random samples
slice_sample(nests_clean, n = 10)

```

```

# A tibble: 10 x 4
  ant_species case_control height_cm full_scientific_name
  <fct>         <dbl>      <dbl> <chr>
1 RRB           0        232  Crematogaster mimosae
2 RRB           0        194. Crematogaster mimosae
3 BBR           0        112  Crematogaster nigriceps
4 RRB           0        368  Crematogaster mimosae
5 BBR           0        140  Crematogaster nigriceps
6 AB            0         85  Crematogaster sjostedti
7 RRB           0        327  Crematogaster mimosae
8 BBR           0        131  Crematogaster nigriceps
9 AB            0        324. Crematogaster sjostedti
10 RRB          0        213  Crematogaster mimosae

```

b. Response Variable

For the `case_control` column, a 1 represents a tree where a bird nest is present while a 0 represents an “available” tree where no birds nests are present.

c. Purpose of Study

Tree height is a continuous variable. As tree height increases the probability of nest occurrence would increase according to paragraph 2 of the results section. The authors say tree architecture has an important role in nest site selection in the second paragraph of the discussion; denser

tree cover likely provides better nest concealment, protecting bird eggs from predators (but nothing about height specifically here).

Acacia ant species is a categorical variable (confirmed in the methods section). The probability of finding a bird's nest is higher for trees occupied by *C. nigriceps* compared to *C. mimosae* (found in second paragraph of results section).

d. Table of Models

4 models total:

Model number	height	ant species	Model Description
1			no interactions (no predictors present)
2	X	X	interaction (height and ant species)

e. Run the Models

```
# model 1: no interaction occurs
# use glm because we want to know how a variable "predicts" or "affects"
# another and the response variable is non-parametric
model1 <-glm(case_control ~ ant_species + height_cm,
             data = nests_clean,
             family = "binomial")# add family = binomial because the
                                # response variable is binary

# model 2: interaction occurs
model2 <-glm(case_control ~ ant_species * height_cm,
             data = nests_clean,
             family = "binomial")
```

f. Select the Best Model

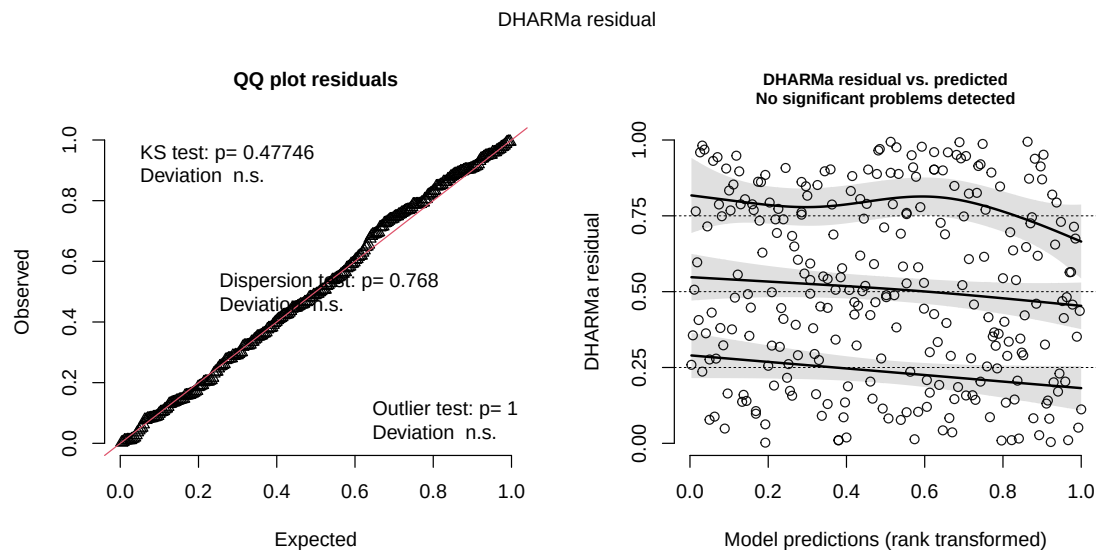
```
# run Akaike's Information Criterion on models
AICc(model1, model2)|>
  arrange(-AICc)
```

	df	AICc
model1	4	258.8940
model2	6	238.1236

The best model as determined by Akaike's Information Criterion (AIC) is the model that suggests there is an interactive effect of ant species and tree height on bird nest occurrence.

g. Check the Diagnostics

```
# use DHARMA package diagnostics for best GLM
plot(simulateResiduals(model2))
```



```
# there is no setting to left-align the variables. I could not figure out how
# to have center text not overlap the line.
```

h. Visualize the Model Predictions

```
# generate predictions to plot
mod_preds <- ggpredict(model2, terms = c("height_cm [all]", "ant_species"))

# base layer ggplot
```

```

ggplot() +

# Raw data points
geom_point(data = nests_clean, # use clean data frame
           aes(x = height_cm, # assign x axis
               y = case_control, #assign y axis
               color = ant_species, #color bt ant species
               group = ant_species,
               alpha = 0.3, size = 2.5)) + # make transparent and visible size

# 95% confidence interval ribbons
geom_ribbon(data = mod_preds,
           aes(x = x, y = predicted,
               ymin = conf.low, ymax = conf.high,
               fill = group),
           alpha = 0.15, inherit.aes = FALSE) +

# Logistic regression prediction lines
geom_line(data = mod_preds,
          aes(x = x, y = predicted, color = group),
          linewidth = 1.2, inherit.aes = FALSE) +

# Scale and presentation formatting
scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, 0.2)) +

# display model predictions for each ant species in a different panel
facet_wrap(~group) +

# Take out grid lines and gray background
theme_minimal() +
  theme(panel.grid = element_blank(),
        text = element_text(face = "bold", size = 11))+

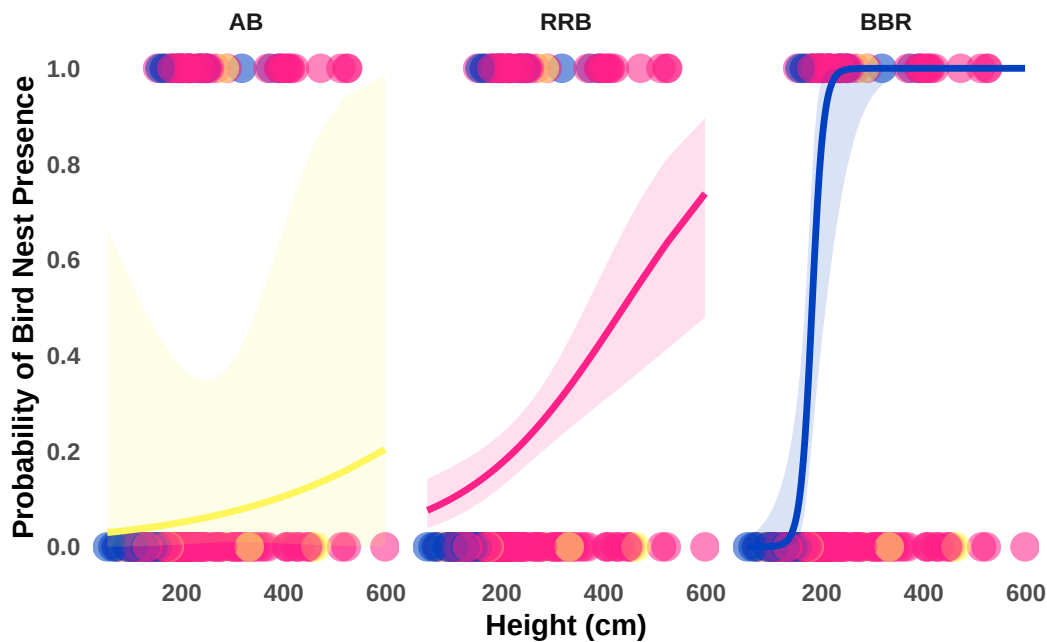
# remove legend
  theme(legend.position = "none")+

# Use colors that are different from the default
scale_color_manual(values = c("#FEF65B", "#FF2088", "#0041C2")) +
scale_fill_manual(values = c("#FEF65B", "#FF2088", "#0041C2")) +

# Add labels with units
labs(

```

```
x = "Height (cm)",
y = "Probability of Bird Nest Presence",
color = "Ant Species",
fill = "Ant Species")
```



i. Write a caption for your figure

Figure 1. The probability of bird nest presence differs depending on tree height and ant species present. The circles represent raw binary observations of bird nest presence (1.0) or absence (0.0) at a specific tree height. The colors represent different ant species occupation of the trees sampled; yellow represents *Crematogaster sjostedti* (AB), pink represents *Crematogaster mimosae* (RRB), blue represents *Crematogaster nigriceps* (BBR). The lines represent the predicted probability of a bird nesting at that corresponding height. The translucent ribbons represent 95% confidence intervals for each species represented. Data from: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.8373322>

j. Calculate Model Predictions

```
# calculate model predictions with model 2 since it was the least bad according
# to AICc
ggpredict(model2,
          terms = c("height_cm [600]", # include height at 600cm as instructed
                    "ant_species")) # include ant species so it displays all 3
```

```
# Predicted probabilities of case_control
```

```
ant_species: AB
```

height_cm	Predicted	95% CI
600	0.20	0.00, 0.99

```
ant_species: RRB
```

height_cm	Predicted	95% CI
600	0.74	0.48, 0.90

```
ant_species: BBR
```

height_cm	Predicted	95% CI
600	1.00	1.00, 1.00

k. Interpret Your Results

Ant species and tree height interact to predict the probability of bird nest occurrence.

Birds are more likely to nest in trees with aggressive defending ant species like *C. nigriceps* (BBR) and *C. mimosae* (RRB), likely because these ants ward off predators. Birds also prefer to nest in taller trees likely because it is harder for land predators to reach their eggs.

Additionally the presence of *C. nigriceps* effects tree structure making birds most likely to choose these trees. In figure 1 the slope of the blue line is steep at ~150 cm suggesting a high probability that birds will nest in BBR trees taller than 150 cm.

In the tallest tree (600 cm), the probability of nesting is 0.20 (95% CI:[0.00,0.99]) for AB, 0.74 (95% CI:[0.48,0.90]) for RRB, and 1.00 (95% CI:[1.00,1.00]) for BBR.

Problem 3. Affective and explanatory visualizations

a. Comparing Visualizations

The visualizations differ in that the exploratory data is more clear and direct where the affective is more artistic and subtle. They are similar in that they represent the same data set. There appears to be a negative relationship between my daily perceived stress level and daily step count in my exploratory data visualization that I chose to display in my affective data visualization. They differ in that the affective visualization does not have any numbers or labels, it is only clear if you read the artist statement. Feedback I got from peers in week 9 included adding information to the artist statement regarding what lines represent data and to reduce clutter meaning I could be less direct and more artistic with incorporating data. I did implement these suggestions by removing raw data points, smoothing the bar graph, and expanding the artist statement.

b. Designing an Analysis

My research question was “Does my daily step count have a positive effect on my stress levels?” I think by positive I meant “good.” To phrase it better: As my daily step count increases, does my daily perceived stress level decrease? To examine the relationship between step count and stress level I could use a Spearman’s correlation. I would use Spearman instead of Pearson because my response variable is ordinal (therefore non-parametric).

c. Check Assumptions and Run Analysis

Check Assumptions of Spearman correlation

Monotonic relationship between variables: Looking at the scatter plot, the data seems roughly monotonic. Independent observations: According to a quick Google search, my data cannot be statistically independent since it all came from me, everyday, my data is inherently linked, but that was the assignment, so I am going to proceed.

Run the Spearman’s Correlation

```
# run spearman correlation test with daily step count and stress level
cor.test(my_data$perceived_stress_level, my_data$daily_step_count,
         method = "spearman")
```

```
Warning in cor.test.default(my_data$perceived_stress_level,
my_data$daily_step_count, : Cannot compute exact p-value with ties
```

Spearman's rank correlation rho

data: my_data\$perceived_stress_level and my_data\$daily_step_count

S = 21329, p-value = 0.005773

alternative hypothesis: true rho is not equal to 0

sample estimates:

rho

-0.4050898

A moderate (0.4) strength relationship is observed.

d. Create Visualization

```
# create blocks out of step values. (Had to look up how to do this because
# my data was looking chaotic.)
my_data_clean <- my_data |>
  mutate(step_groups = cut(daily_step_count,
                           breaks = seq(0, 20000, by = 2000),
                           labels = c("0-2k", "2k-4k", "4k-6k", "6k-8k", "8k-10k",
                                       "10k-12k", "12k-14k", "14k-16k", "16k-18k", "18k-20k"),
                           include.lowest = TRUE))

# calculate medians to represent Spearman correlation
medians <- my_data_clean |>
  group_by(step_groups) |>
  summarize(median_stress = median(perceived_stress_level))

# create plot, assign x and y values
# X is my discrete predictor variable, Y is ordinal response variable.
ggplot(my_data_clean, aes(x = step_groups, y = perceived_stress_level))+

# Raw data points jittered to prevent overlap
geom_jitter(
  shape = 19,
  size = 3,
  alpha = 0.4,
  color = "blue",
  width = 0.2,
```

```

    height = 0.15
  ) +

# plot median points
geom_point(
  data = medians,
  aes(x = step_groups, y = median_stress),
  color = "#d94801",
  size = 3
) +

# plot median trendline
geom_line(
  data = medians,
  aes(x = step_groups, y = median_stress, group = 1),
  color = "#d94801",
  linewidth = 1.2
) +

# graph title and axis titles
labs(title = "Perceived Stress by Daily Step Count",
      subtitle = "6/3/2026",
x = "Daily Step Count (steps)",
y = "Perceived Daily Stress Level (1-5)")+

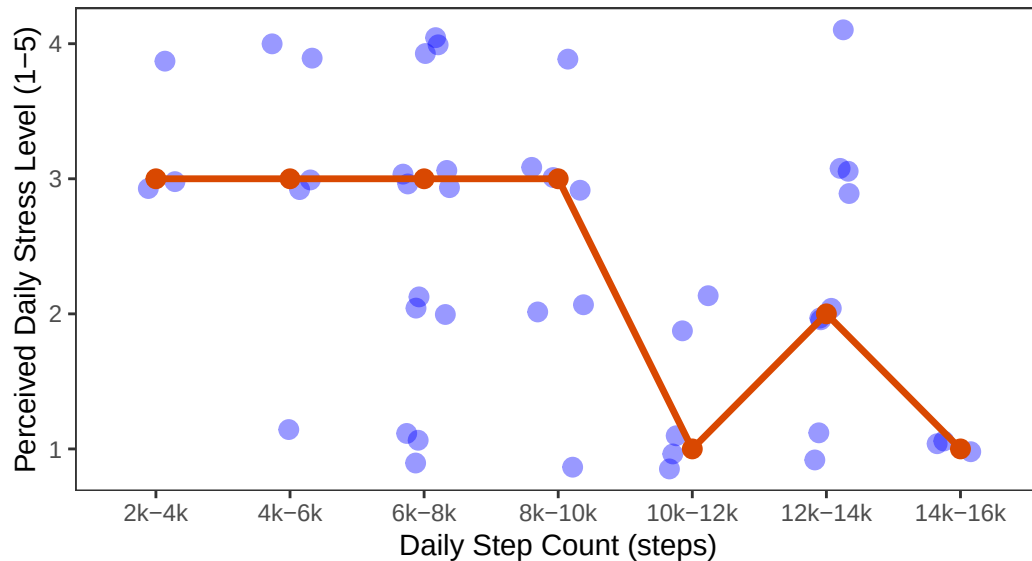
# choosing non-default theme
theme_bw()+

# remove grid
theme(panel.grid = element_blank())

```

Perceived Stress by Daily Step Count

6/3/2026



e. Write a caption

Figure 2. Higher step count tends to correlate with a lower stress level. Each blue dot represents one days total step count and stress level. Step counts were grouped into bins every 2,000 steps. Each red dot represents the median stress level for that step count bin. The red line is a median trend line.

f. Write about results

I found a moderate negative relationship between daily step count and daily perceived stress level (Spearman $\rho = -0.41$, $S = 21329$, $p < 0.006$, $\alpha = 0.05$). My results suggest that as daily step count increases stress level decreases.

g. Sharing your affective visualization

Shared in week 10 workshop.